

AIM PROTOCOL

Agri • Identity • Microfinance

Version 2.3.1 — Release Notes & Technical Reference

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Post-V2.3 Stabilisation — July 2026

Live: <https://aim-protocol.vercel.app/>

Status	Live on Solana Devnet
Frontend Repo	github.com/amjadkamara/aim-protocol
Smart Contract Repo	github.com/amjadkamara/aim-program
Program ID	AhHHJTU5vodDYE2yLNet2bE6jad9F3xSfbLQdUmykKqB
Program ID Change	None — upgrades the existing deployment, no redeploy to a new address

1. Overview

V2.3 shipped the multi-stakeholder frontend — the public-facing marketplace, lender registration, and farmer profile lookup pages described in the V2.3 release notes. V2.3.1 is a stabilisation pass on top of that: it closes two correctness bugs that were silently breaking the core loan lifecycle, and rebuilds the app's visual language to match the AIM Protocol Product Design System v1.0 — flat, restrained, single-accent, enterprise-grade rather than hackathon-grade.

This version does not introduce new product surface area. It makes the existing surface actually work end to end, and makes it look like infrastructure a lender or regulator would trust.

2. What V2.3.1 Fixed

2.1 Loan Request — Missing Lender Selection (Critical)

File: `src/Microloan.jsx`, `src/useAimProgram.js`

`requestLoan()` in `useAimProgram.js` requires a `lenderPubkey` account — the loan is tied to a specific lender's on-chain terms, not a generic pool. `Microloan.jsx`'s form never collected one, so every loan request sent `lender: undefined` to Anchor and failed before it ever reached the chain, with no specific error surfaced to the farmer beyond a generic "Transaction failed." This made loan requests completely non-functional.

Fix: `Microloan.jsx` now fetches all active lenders and presents them as a required first step — flat selectable cards showing organisation name, max loan, APR, and duration. Once a lender is picked, the amount and repayment-period fields constrain themselves to that lender's declared terms, so a farmer cannot request more than a lender actually offers. The selected lender's public key is passed through to `requestLoan()`.

2.2 Repaid Loans Disappearing From the Record (Critical)

File: `programs/aim-program/src/lib.rs`

Two compounding bugs in the `repay_loan` instruction:

- `repay_loan()` never set `loan.is_repaid = true` — the flag stayed false even after a successful repayment.
- The `RepayLoan` account context had `close = owner` on the loan account, closing it and reclaiming rent automatically the moment repayment succeeded.

Net effect: a repaid loan's on-chain account was destroyed before it was ever correctly flagged repaid. The farmer's own dashboard showed "no active loan" immediately post-repayment with no confirmation, and the Admin Dashboard's Repaid stat was permanently stuck at 0%, because `getProgramAccounts` can't return an account that no longer exists. This also made the existing `close_loan` instruction — and the "Clear repaid loan account to re-borrow" button already built in `Dashboard.jsx` for it — unreachable dead code, since `repay_loan` was closing the account first.

Fix: `repay_loan()` now sets `is_repaid = true` and no longer closes the account. The loan stays on-chain, correctly flagged, until the farmer explicitly clears it via `close_loan` — restoring the lifecycle the frontend was already built to expect. No account layout or space changes; deployed as a same-Program-ID upgrade, no new address, no frontend changes required for this fix specifically.

2.3 Admin Dashboard — Loan Status Never Resolving (Critical)

File: `src/AdminDashboard.jsx`

`getLoanStatus()` in `AdminDashboard.jsx` assumed `LoanAccount` had a Rust enum status field (`Object.keys(loan.status)[0]`). The actual on-chain shape — confirmed by the working implementation already in `Dashboard.jsx` — is a plain `isRepaid: bool` plus `createdAt/repaymentWeeks`, with status computed client-side from those. `loan.status` doesn't exist on the real account, so the function silently returned null for every loan: blank status badges, all four stat cards stuck at zero, and every status filter broken (everything defaulted into the "no loan" bucket internally).

Fix: replaced with the same time-based calculation already proven correct in `Dashboard.jsx`. All existing call sites (stat counts, sort, filter) already passed the full loan object rather than `loan.status`, so this was a contained fix to the one function plus its badge component.

2.4 Pre-Connection Admin Routing (Logic)

Files: `src/FarmerID.jsx`, `src/LenderRegistration.jsx`

The admin wallet is blocked from both Farmer and Lender roles at the contract level, but the UI didn't check for it on arrival at either registration page. Clicking "Register as Lender" or "For farmers" before connecting a wallet, then connecting as admin from the gate screen, dropped the admin wallet into the registration form instead of routing it home. Fixed with an admin-wallet check that redirects home the instant the connected wallet is detected as admin, covering both the already-connected and connects-mid-flow cases.

Separately, `FarmerID.jsx` had no wallet-required gate at all — it rendered the full registration form even with no wallet connected, with `handleSubmit` silently no-op'ing on click. Added the same wallet-required gate already used in `LenderRegistration.jsx`.

3. Enterprise UI Pass

Applied across `Layout.jsx`, `App.jsx`, `Dashboard.jsx`, `Microloan.jsx`, and `AdminDashboard.jsx`, per the AIM Protocol Product Design System v1.0:

- Removed all hackathon/competition references and emoji from the live product (footer, stats bar, hero badge).
- Removed decorative gradient/glow effects — background glow orbs, gradient button fills, gradient card accents, gradient CTA overlays.
- Replaced unicode status glyphs (●, ○, ◐, △, ✓) with a shared, flat, bordered `StatusBadge` component across every screen that shows loan or lender status.

- Consolidated the accent palette to a single restrained green, with color reserved for genuine status meaning (green = active/repaid/approved, amber = pending, red = overdue) rather than decorative variety on headers and primary buttons.
- Unified status colors that had drifted inconsistent across files — Dashboard.jsx used amber for "active loan" while AdminDashboard.jsx used blue for the same state; lender "pending" was yellow in one file, amber in another. Now consistent everywhere.
- Sentence case throughout — no more ALL CAPS field labels or button text.
- Rebuilt Layout.jsx's nav to match the design system's structure (Protocol / For farmers / For lenders / Docs) and simplified the footer.
- Rebuilt App.jsx's hero to the calmer, smaller type scale and copy tone specified in the design system, replacing the original hackathon-era hero.
- Removed the landing page stats bar per direct instruction — flagged that this may have been serving a legitimate "Trust Strip" role per the design system's 11-section structure, and left as an open decision for the next version.

4. File Changelog

File	Status	What Changed
src/Microloan.jsx	Updated	Added required lender selection; amount/duration now constrained to selected lender's terms; wired lenderPubkey into requestLoan(); flat enterprise visual pass
src/useAimProgram.js	Reviewed	requestLoan()'s existing lenderPubkey requirement confirmed as the root cause of silent loan failures; no changes needed here — fix was in the calling component
programs/aim-program/src/lib.rs	Updated	repay_loan now sets is_repaid = true; close = owner removed from RepayLoan account context; close_loan is now reachable as intended
src/AdminDashboard.jsx	Updated	Fixed getLoanStatus() to match the real LoanAccount shape instead of a nonexistent status enum; flat enterprise visual pass; unified status colors with rest of app
src/FarmerID.jsx	Updated	Added wallet-required gate (previously rendered the form with no wallet connected); added admin-wallet redirect
src/LenderRegistration.jsx	Updated	Added admin-wallet redirect on connect, covering both already-connected and connects-mid-flow cases
src/LoanMarketplace.jsx	Updated	Fixed reading maxLoanLamports instead of stale maxLoanAmount field name (V2.3 contract renamed this field)
src/Layout.jsx	Updated	Rebuilt nav and footer per design system; removed background glow effects and hackathon references
src/App.jsx	Updated	Rebuilt hero and home states per design system; removed stats bar and gradient effects; parallelised farmer/lender role-resolution fetch

src/Dashboard.jsx	Updated	Parallelised lender/farmer/loan fetch; flat status badges; sentence case; single accent color
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5. Verification Checklist

Loan Lifecycle

- ✓ Farmer can select an active lender and request a loan within that lender's terms
- ✓ Loan request succeeds on-chain (previously failed silently on every attempt)
- ✓ Repaid loan remains visible and correctly flagged after repayment
- ✓ Admin dashboard shows accurate Active/Overdue/Repaid counts and repaid rate
- ✓ Admin status filters (Active/Overdue/Repaid/No loan) correctly isolate matching farmers
- ✓ "Clear repaid loan account to re-borrow" button is now reachable and functional

Wallet & Routing

- ✓ FarmerID.jsx gates behind wallet connection instead of rendering the form pre-connection
- ✓ Admin wallet redirected home from both FarmerID.jsx and LenderRegistration.jsx, pre- and mid-connection
- ✓ Nav links (Protocol / For farmers / For lenders / Docs) route correctly

Visual

- ✓ No hackathon references or emoji on the live product
- ✓ No decorative gradients, glows, or unicode status glyphs remain in the files covered by this pass
- ✓ Status colors consistent across Dashboard.jsx, AdminDashboard.jsx, and Microloan.jsx

6. Known Open Items — Next Version

Not deferred by accident — surfaced during this pass and consciously scoped out to keep V2.3.1 focused on correctness and core visual consistency.

- LoanMarketplace.jsx has not received the enterprise visual pass — still on the original yellow/ALL CAPS treatment.
- RepayLoan.jsx itself was never opened this version; the repay_loan contract fix means Dashboard.jsx now shows a correct repaid state, but RepayLoan.jsx's own confirmation screen should be reviewed directly.
- No farmer detail modal in AdminDashboard.jsx — lenders have one (name, terms, wallet, approve action), farmers do not.
- reject_lender instruction and the corresponding "Reject" admin action — LenderAccount currently has no way to represent "reviewed and declined," only pending/active.

Deliberately deferred alongside lender approval email notifications, both flagged for a dedicated pass.

- Full landing page rebuild against the AIM Protocol Product Design System v1.0's 11-section structure (Hero, Trust Strip, Protocol Overview, How it Works, For Farmers, For Lenders, Developer Platform, Marketplace, Security, CTA, Footer) — current build covers roughly 4 of 11 sections. Open decision on whether the removed stats bar should return in redesigned form as the design system's "Trust Strip."
- Tiered lending model (trust-ladder pools, on-chain reputation scoring, escrow-based disbursement) remains explicitly out of scope — a locked v1 roadmap document, not part of the current contract or UI.

AIM Protocol V2.3.1 — Loan lifecycle correctness restored. Enterprise UI pass on core screens complete.

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